Start with the debt the issuer promises

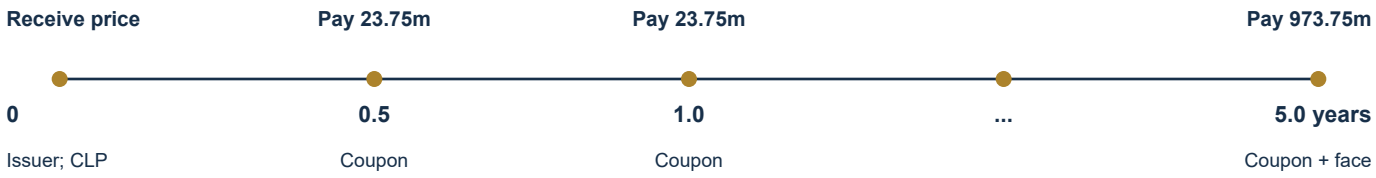
LECTURE 8 · REVIEW NOTES

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Suppose Sierra Copper, a hypothetical U.S. mining company, issues debt in Chile. I will follow the transaction twice: first in pesos, then in dollars. The company sets the local face equivalent at USD 1 million. With USDCLP at t=0 equal to 950 CLP per USD, face value is CLP 950 million. The promised payments remain in pesos even though the CFO evaluates the financing in dollars.

Bond term	Value
Face value	CLP 950,000,000
Annual coupon	5.00%
Maturity and payments	5 years; semiannual
Coupon every six months	CLP 23,750,000
Final coupon plus principal	CLP 973,750,000

All rates in this example are annual nominal rates compounded semiannually. The issuer receives the bond price today and pays ten coupons plus principal. The investor has exactly the opposite cash-flow signs.



Face equivalent is not issue proceeds

Face value is defined using today's spot quote, but actual proceeds depend on the bond price. If the local required yield exceeds the 5% coupon, the bond sells below face and the issuer receives less than the spot equivalent of USD 1 million.

$$P^{\text{spot}}_{\text{USD},0} = \frac{P_{\text{CLP},0}}{\text{USDCLP}_0}$$

Read USDCLP in the correct direction

USDCLP means CLP per USD. Divide pesos by USDCLP to obtain dollars. A higher USDCLP means a weaker peso and fewer dollars needed to pay a fixed peso obligation.

Value the local bond and summarize its yield

Discount each peso payment using the local zero rate for its maturity. Then solve for the single YTM that reproduces that price. Curve-based valuation and yield interpretation are two distinct steps.

$$P_{\mathrm{CLP},0} = \sum_{i=1}^{10} \frac{CF_{\mathrm{CLP},t_i}}{(1 + r_{\mathrm{CLP},t_i}/2)^{2t_i}}$$

Years	CLP zero rate	USD zero rate	Forward USDCLP
0.5	5.50%	4.10%	956.5164
1.0	5.55%	3.95%	964.9641
1.5	5.60%	3.85%	974.6772
2.0	5.65%	3.80%	984.9671
2.5	5.70%	3.78%	995.6055
3.0	5.75%	3.75%	1,007.3421
3.5	5.80%	3.72%	1,020.0126
4.0	5.85%	3.70%	1,033.2427
4.5	5.90%	3.68%	1,047.3582
5.0	5.95%	3.65%	1,062.9123

The local price is CLP 912,284,205.32, with annual nominal YTM 5.9291%. Price is below face and YTM exceeds coupon: the bond trades at a discount. Converting the issue price at today's spot gives USD 960,299.16 of proceeds. This does not yet fix the dollar cost of future peso payments.

Use the annuity formula as a check

Once the local YTM is known, the Lec07 coupon-annuity formula plus the present value of principal reproduces the price. In Excel, 2 x RATE(10, 23750000, -price, 950000000) reports the local annual nominal YTM.

These are illustrative curves

The rates are teaching inputs, not current market quotes. The forward column is calculated from both curves and the 950 spot rate.

Hedge the payments and value them in dollars

A matching-maturity FX forward locks the dollar cost of each peso payment. Because USDCLP is pesos per dollar, covered interest parity places the CLP growth factor in the numerator and the USD growth factor in the denominator.

$$\mathrm{USDCLP}^{F}_{0,t} = \mathrm{USDCLP}_0 \frac{(1 + r_{\mathrm{CLP},t}/2)^{2t}}{(1 + r_{\mathrm{USD},t}/2)^{2t}}$$

$$CF^{H}_{\mathrm{USD},t} = \frac{CF_{\mathrm{CLP},t}}{\mathrm{USDCLP}^{F}_{0,t}}, \qquad P^{H}_{\mathrm{USD},0} = \sum_t \frac{CF^{H}_{\mathrm{USD},t}}{(1 + r_{\mathrm{USD},t}/2)^{2t}}$$

The first forward is 956.5164, making the first coupon a locked USD 24,829.68 payment. The last forward is 1062.9123; the final coupon plus principal becomes USD 916,115.11. Each payment uses its own forward and USD discount rate.

The price identity is the economic result

Substitute the parity forward into the USD present-value formula. The USD growth factor cancels. Each dollar present value equals the corresponding local present value divided by spot. Adding all dates gives:

$$P^{H}_{\mathrm{USD},0} = \frac{P_{\mathrm{CLP},0}}{\mathrm{USDCLP}_0}$$

Both methods give USD 960,299.16. The local YTM is 5.9291%, while the hedged dollar cash-flow YTM is approximately 3.6619%. The lower dollar yield is not a free funding advantage: the dollar cash-flow amounts also differ.

Why RATE is not generally exact after conversion

Peso coupons are level, but dollar coupons vary with the forwards. Solve the USD pricing equation for one yield, or use twice the periodic IRR with initial proceeds and all ten payments. A fixed-payment RATE calculation would change the cash-flow pattern.

Compare the issuer's costs across FX scenarios

After issuance, initial proceeds are fixed. Without a hedge, divide each peso payment by spot USDCLP on its payment date. With a hedge, use the already locked forward. Discount both payment streams with the same USD curve.

$$\text{Savings}_0 = PV_{\text{USD}}(\text{unhedged payments}) - PV_{\text{USD}}(\text{hedged payments})$$

For a transparent comparison, assume a constant future USDCLP across payment dates. Today's issuance spot remains 950 in every case. Only the future conversion path changes.

Future USDCLP	Unhedged cost PV	Hedged cost PV	Hedge savings
760	1,325,832.64	960,299.16	365,533.48
950	1,060,666.11	960,299.16	100,366.95
1,140	883,888.43	960,299.16	-76,410.74

At 760, the stronger peso requires more dollars to service the debt; the hedge protects the issuer. At 1,140, the weaker peso makes unhedged payments cheaper, so hedge savings are negative. This is the opportunity cost of fixing the exchange rates. The hedge still delivers predictable dollar payments.

From currency risk to interest-rate risk

An FX forward fixes the conversion rate for a future currency payment. A forward rate agreement fixes an interest-rate exposure for a future borrowing or investment period. The uncertainty changes, but the logic is similar: combine the underlying transaction with an offsetting derivative.

A company will borrow USD 1 million for six months beginning six months from now. Locking a 4% annual semiannual-compounded rate fixes interest at USD 20,000. If the market rate at borrowing is 6%, interest is USD 30,000 and a USD 10,000 end-date payoff offsets the increase. Lec09 develops this timing and both sides of the hedge.

Compare costs, not isolated derivative profits

A positive saving means a lower issuer cost relative to remaining unhedged. It is not an additional issue proceed and should not be added to the bond price.